

4. Round Robin Scheduling

C Program :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, tq, i;
```

```
    int bt[20], rem_bt[20];
```

```
    int wt[20] = {0}, tat[20];
```

```
    int time = 0, done;
```

```
    float avg_wt = 0, avg_tat = 0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 0; i < n; i++) {
```

```
        printf("Enter Burst Time for P%d: ", i + 1);
```

```
        scanf("%d", &bt[i]);
```

```
        rem_bt[i] = bt[i];
```

```
    }
```

```
    printf("Enter Time Quantum: ");
```

```
    scanf("%d", &tq);
```

```
    do {
```

```
        done = 1;
```

```
        for (i = 0; i < n; i++) {
```

```
            if (rem_bt[i] > 0) {
```

```
                done = 0;
```

```
                if (rem_bt[i] > tq)
```

```
                {
```

```
                    time += tq;
```

```
                    rem_bt[i] -= tq;
```

```
                }
```

```

        else {
            time += rem_bt[i];
            wt[i] = time - bt[i];
            rem_bt[i] = 0;
        }
    }
}

} while (!done);

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    tat[i] = bt[i] + wt[i];
    avg_wt += wt[i];
    avg_tat += tat[i];
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

avg_wt /= n;
avg_tat /= n;

printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

return 0;
}

```

Shell Program (rr.sh) :

```

#!/bin/bash

echo "Enter Number of Processes"

read n

for ((i = 0; i < n; i++))
do
    echo "Enter Burst Time for P$((i + 1))"
    read bt[$i]

```

```

    rem_bt[$i]=$(bt[$i])
done
echo "Enter Time Quantum"
read tq
time=0
while true
do
    done=1
    for ((i = 0; i < n; i++))
    do
        if [ ${rem_bt[i]} -gt 0 ]
        then
            done=0
            if [ ${rem_bt[i]} -gt $tq ]
            then
                time=$((time + tq))
                rem_bt[$i]=$((rem_bt[i] - tq))
            else
                time=$((time + rem_bt[i]))
                wt[$i]=$((time - bt[i]))
                rem_bt[$i]=0
            fi
        fi
    done
    [ $done -eq 1 ] && break
done

echo
echo -e "Process\tBT\tWT\tTAT"

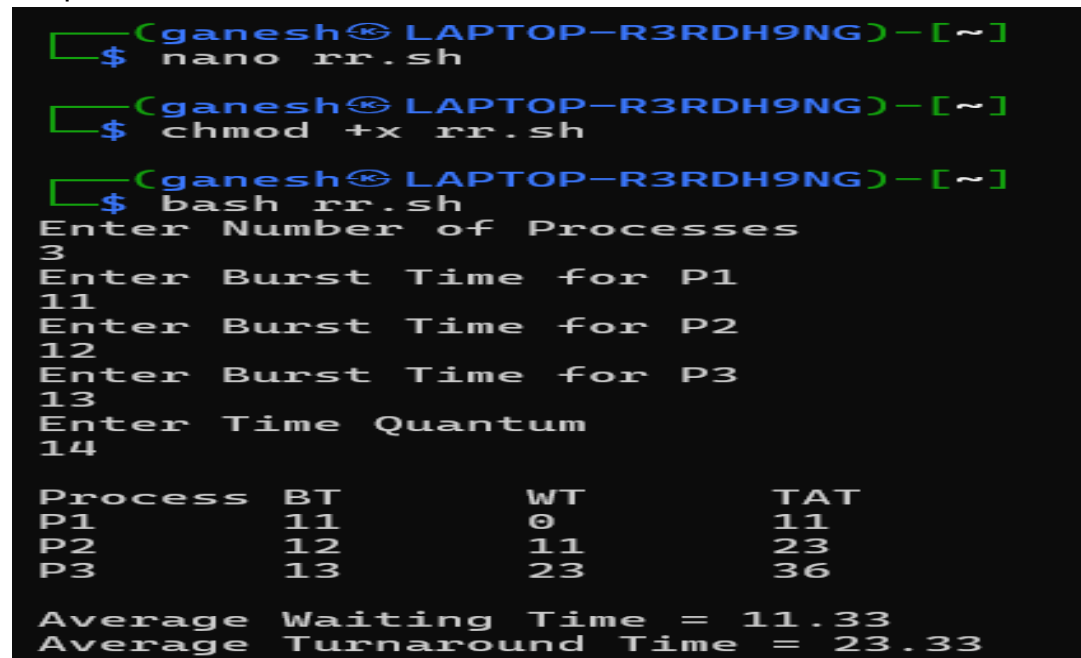
```

```

total_wt=0
total_tat=0
for ((i = 0; i < n; i++))
do
    tat[$i]=$((bt[i] + wt[i]))
    total_wt=$((total_wt + wt[i]))
    total_tat=$((total_tat + tat[i]))
    echo -e "P$(i + 1)\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt / $n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat / $n}")
echo
echo "Average Waiting Time = $avg_wt"
echo "Average Turnaround Time = $avg_tat"

```

Output :



```

(ganesh@LAPTOP-R3RDH9NG)~$ nano rr.sh
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x rr.sh
(ganesh@LAPTOP-R3RDH9NG)~$ bash rr.sh
Enter Number of Processes
3
Enter Burst Time for P1
11
Enter Burst Time for P2
12
Enter Burst Time for P3
13
Enter Time Quantum
14

Process  BT      WT      TAT
P1       11       0       11
P2       12       11      23
P3       13       23      36

Average Waiting Time = 11.33
Average Turnaround Time = 23.33

```